

Workshop 2

Welcome!

Find a group to work with & say hi!

ECO 101: Principles of Microeconomics

Workshop Overview

- Opportunity Cost
- Patterns of Trade
- Production Possibility Frontiers (PPFs)

Opportunity Cost

P1: what does each of you give up?

Setting: Consulting Firm

- Two tasks: client calls and rows of data entry
- Two workers: you and an AI agent "AIDAN"

In one hour	Client calls	Data entry rows
AIDAN	4	40
You	2	10

Three Principles: Your perspective

Step 1: What is the choice?

- Choice:

P1: Benefit vs Opportunity Cost

- Benefit:
- Opp. Cost:
 - Resources:
 - Next best alternative:
- OC of 1 call:

What does one call cost?

- Cost units: Rows of data entry
- Cost is expressed in units of the other thing you could have produced instead with the same resources
- Last week's thinking and equation, with different units

Opportunity Cost of 1 call

In one hour	Client calls	Data entry rows
AIDAN	4	40
You	2	10

- You:
- AIDAN:

Firm needs 16 client calls a day

- 16 calls non-negotiable. Extra time goes into data entry.
- Two employees, each with exactly 8 hours a day. Same productivity as before

What should the firm do?

1. Use only AIDAN, fire you
2. AIDAN: 16 calls + data entry, you: data entry
3. You: 16 calls + data entry, AIDAN: data entry

Worksheet Q1 (counts for grades)

Put answers on worksheet

Summary

- Absolute advantage: who can produce more with 1 unit of the resource
- Comparative advantage: who gives up less of the other good to produce 1 unit
- Opportunity Cost (OC) captures production trade-offs from reallocating resources
 - OC depends on *relative* productivity
 - Expressed in units of the other good
 - Only relevant when all resources are being put to use efficiently

Patterns of Trade

Knowing who should trade what is not the same as knowing on what terms

Two independent contractors

- Two contractors, each with 8 hours of time per day
- One sells customer calls and receives rows of data entry in return

Two questions:

- Who should sell calls? Why?
- What minimum number of data rows per call makes them agree to the trade?

Two independent contractors

In one hour	Client calls	Data entry rows
AIDAN	4	40
You	2	10

- Who should sell calls? Why?
- What minimum number of data rows per call makes them agree to the trade?

Trade if...

- ...What you're offered is at least as good as your next best option
- Here: Next best option is doing it yourself at your Opp. Cost of production
- Sell if you're offered at minimum your OC of producing it
- Buy if you're paying at most your OC of producing it yourself.
- *Same thinking as last week: keep the thing you care about on one side, push the rest across.*

Summary

- Trade patterns depend on comparative advantage
- Incentives to trade depend on the price (in units of the good) or the exchange rate
- Both better off with trade if exchange rate lies between the buyer's Opp. Cost and the seller's Opp. Cost

Production Possibility Frontiers

Can we get more without losing anything by reallocating resources?

Two people, two goods

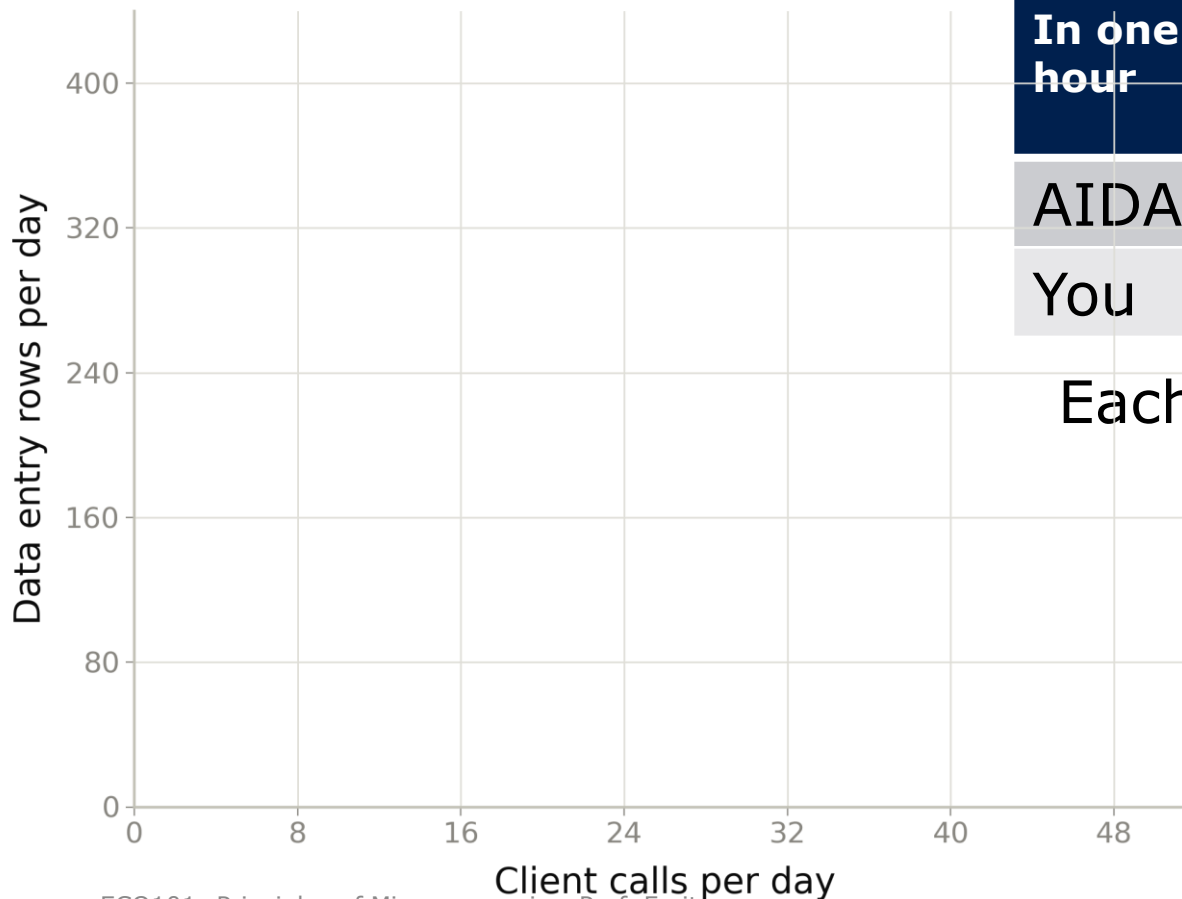
Typical questions

- How to allocate people's finite resources to good production?
- Is this allocation efficient?
 - Efficient if you cannot get more of (at least) one good without giving up some of the other good
- PPF: Production Possibility Frontier
 - Helps us answer the questions quickly using a graph

Individual PPFs

PPF: Production Possibility Frontier

Draw clearly labeled PPFs for you and AIDAN



In one hour	Client calls	Data entry rows
AIDAN	4	40
You	2	10

Each has 8 hours of time

Joint PPF: Working together

PPF: Production Possibility Frontier

- Points on the boundary: maximum that can be produced with the given resources

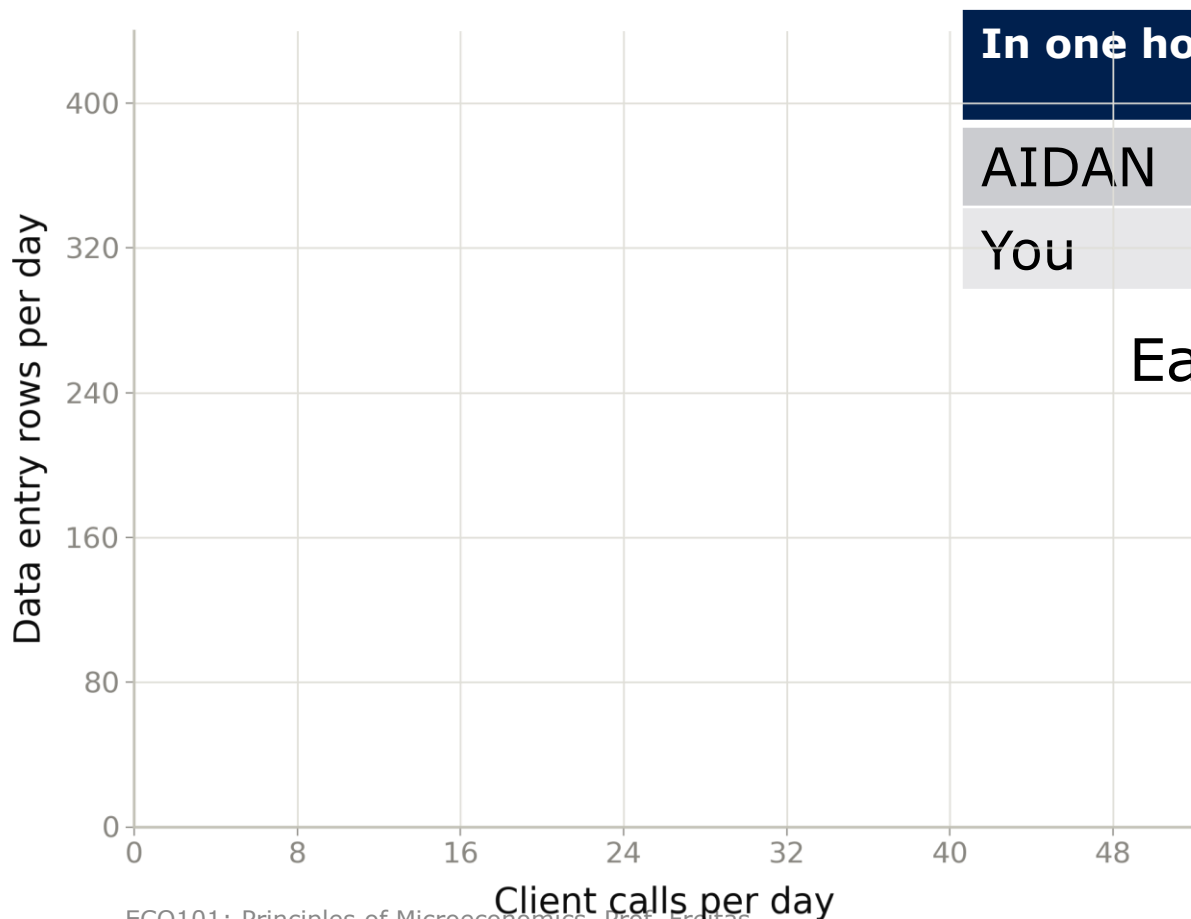
Process:

- Start with everyone's resources in one good
- Produce the first unit of other good by moving the lowest cost producer into its production
- If they run out of time, switch to the next lowest cost person. Continue.
- Stop when everyone has all their time in the other good's production

Joint PPF

PPF: Production Possibility Frontier

Draw clearly labeled joint PPF



In one hour	Client calls	Data entry rows
AIDAN	4	40
You	2	10

Each has 8 hours of time

	OC of 1 call
AIDAN	10 rows
You	5 rows

Worksheet Q2 (counts for grades)

Put answers on worksheet

Something to think about at home

Don't forget to give me your
completed worksheets

See you next workshop!
